CLAUDIA MANNA, PhD

R&D SCIENTIST | COSMETIC FORMULATION & PRODUCT DEVELOPMENT | ORGANIC CHEMISTRY

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PROFESSIONAL SUMMARY

From making molecules to making products, my path has gradually moved from organic chemistry and pharmaceutical science toward cosmetic formulation, ingredient innovation, and new product concepts. With a PhD in Organic Chemistry and current training in Cosmetic Chemistry at UC Riverside, I am building on my scientific background to move into formulation-focused R&D.

CORE SKILLS

FORMULATION & PRODUCT DEVELOPMENT

- O/W and W/O emulsions
- Creams, gels, cleansing systems
- Raw-material selection
- Stability, pH and viscosity
- Sensory evaluation
- Benchmarking and product briefs
- The Good Face Formulator

ORGANIC CHEMISTRY & R&D

- Multistep synthesis
- Reaction optimization
- Stereochemistry
- Purification
- Troubleshooting
- Structure-property relationships

ANALYTICAL CHARACTERIZATION

- NMR (1D/2D)
- HPLC; LC-MS/MS
- GC; IR; UV-Vis
- Data interpretation
- Purity assessment
- Method development

SCIENTIFIC & PRODUCT KNOWLEDGE

- GMP principles
- Skin and hair biology
- Cosmetic labeling and claims
- Manufacturing fundamentals

COSMETIC SCIENCE | FORMULATION TRAINING

CURRENT TRAINING 2026

Cosmetic Chemistry - Formulation Lab Training

UC Riverside Extension | California

- Developed an original skincare concept from brief to lab prototype, selecting raw materials based on functionality, compatibility, sensory goals, and target performance.
- Prepared and evaluated O/W emulsions, moisturizing systems, gels, and surfactant-based cleansers; assessed pH, viscosity, sensory properties, and preliminary stability.
- Applied benchmarking, ingredient research, packaging considerations, and structured documentation to identify formulation improvements and next steps.
- Used *The Good Face Formulator* to evaluate formulations and ingredient compatibility.

2008-2016

Pharmaceutical & Cosmetic Formulation

University of Naples Federico II | Italy

- Formulated creams, ointments, gels, solutions, and O/W and W/O emulsions; applied pharmaceutical formulation principles to excipient selection, pH, texture, and basic stability.

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PROFESSIONAL EXPERIENCE

TEACH

Santa Clara University |California

Lecturer, General & Organic Chemistry
2025-2026

- Designed and delivered undergraduate lectures in General and Organic Chemistry, translating complex chemical concepts into clear, accessible explanations.
- Guided students through laboratory methods, data interpretation, experimental troubleshooting, and safe laboratory practice.

CARE

Community Pharmacy | Italy

Pharmacy Assistant
2013-2016

- Supported customer education on OTC and dermocosmetic products, translating technical product information into clear, practical language.

PhD

Otto-von-Guericke University | Germany

Doctoral Researcher - Organic Chemistry
2020-2024

- Led complex multistep organic synthesis projects from experimental planning and reaction optimization through purification and full compound characterization.
- Resolved selectivity and stereochemical challenges using systematic condition screening, 1D/2D NMR interpretation, chromatography, mass spectrometry, and complementary analytical data.
- Developed reproducible experimental procedures and technical documentation while collaborating across scientific teams; contributed to patented work on novel neurotrophic terpenes.

ANALYSIS

IDAEA-CSIC | Spain

Analytical Chemistry Researcher
2016

- Performed analytical characterization using LC-MS, GC, and spectroscopic techniques, interpreting data for compound identification and purity assessment.

R&D

Otto-von-Guericke University | Germany

Organic Chemistry Research Scientist
2018-2019

- Conducted experimental research on complex organic molecules, supporting synthesis, reaction optimization, analytical interpretation, and detailed laboratory documentation.

MED-CHEM

University of Naples Federico II |Italy

Medicinal Chemistry Intern
2015-2016

- Contributed to a medicinal chemistry project by synthesizing and characterizing anthraquinone-based small molecules investigated for potential antiviral activity against the HIV-1 nucleocapsid protein.

What this experience adds to formulation R&D

SCIENTIFIC FOUNDATION

Molecular understanding, structure–property relationships, and raw-material knowledge.

R&D PROBLEM-SOLVING

Experimental design, optimization, troubleshooting, and data-driven decisions.

CURIOSITY & GROWTH MINDSET

Open to new methods, eager to learn, and ready to bring scientific skills where they add value.

COLLABORATION & COMMUNICATION

Cross-functional communication, scientific storytelling, and the ability to translate technical concepts clearly.

EDUCATION

CERT — *UC Riverside Extension | California*

Expected November 2026

Professional Certificate in Cosmetic Chemistry

MSc — *University of Naples Federico II, Italy*

July 2019

Drug Biotechnologies

PhD — *OVGU University | Germany*

December 2024

PhD, Organic Chemistry

MSc — *University of Naples Federico II, Italy*

March 2016

Chemistry and Pharmaceutical Technology-Pharmacy

INNOVATION & SELECTED PROJECTS

FOUNDER & CREATOR

- ***Molekül Archiv*** *[MÖ-le-kül ar-KEEV]* | Independent science-and-culture blog — **Launching 2027**

A personal editorial project using molecules as entry points to explore how chemistry shapes medicine, cosmetics, and food.

Explore the project → claudiamanna.com

BEYOND THE LAB



Food, Wine & Culture

Cooking, baking, sharing traditions, and bringing people together around the table.



Science Storytelling

Turning technical ideas into clear, engaging stories is one of the parts of science I enjoy most.



Theatre

An expressive space that shaped confidence, communication, and curiosity.



Swimming

My favorite sport and a place for focus, rhythm, and balance.



Creativity & Design

I grew up drawing and handcrafting leather bags with my mother, a handbag designer.

A SMALL PERSONAL DETAIL

I grew up in a seaside town in Southern Italy, but fall has always felt special to me...
The first cool breeze still brings that feeling of a fresh start.

LANGUAGES

Italian	English	Spanish	German
Native	Fluent	Fluent	Intermediate